

MA3PPR - Individual Report Project Catalogue

This catalogue contains project descriptions for the MA3PPR Individual Report Project. This document should be read in conjunction with the Individual Report Project Description document, and taking into account information supplied in the lectures for the module.

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1 Regression models for spectroscopy data

DR JULIA ABERY

1.1 Overview

Near infrared spectroscopy (NIR) and mid infrared spectroscopy (MIR) are methods used routinely in chemometrics, where absorption of near or mid-infrared light is measured for a sample at various wavelengths. Where information is available on single or multiple response variables (typically other measurements taken on the sample, possibly obtained by 'wet' chemistry), the data are naturally suited to regression analysis. However, the large number of correlated explanatory variables means that basic multiple linear regression may not be appropriate, because of multicollinearity, and other approaches may be needed.

For this project, you will investigate the issue of multicollinearity in standard linear regression models. You will then explore a number of modern regression methods which provide an alternative approach. You will conduct a comparative evaluation of the different methods by applying them to spectroscopy datasets.

Students can use R or another statistical computing package of their choice to fit and assess the models.

1.2 Link to global sustainability

Spectroscopy and modern regression methods are applied in environmental contexts such as soil health monitoring and carbon sequestration.

1.3 Pre-requisites

The project builds on topics from linear models (ST2LM / ST2LMD) and machine learning (ST3MML). Experience of programming is also required (e.g. from MA1MSP, CS1PC20 or from elsewhere).

1.4 Questions

Question 1 [10 marks]

Multicollinearity is a problem when analysing spectroscopy data. Write code which enables you to demonstrate the problems that multicollinearity causes with the estimation of parameters in a multiple linear regression model. Choose how to best represent your results and give a full explanation of your findings.

Question 2 [15 marks]

Using information you find in the literature, and from your studies in ST3MML, describe in detail the methods of principal components regression, partial least squares regression and ridge regression. Include a summary of their advantages and disadvantages.

Question 3 [25 marks]

Apply each of principal components regression, partial least squares regression and ridge regression, as well as standard multiple regression, to your dataset, and compare and interpret the results.

Question 4 (MA3PPNR students only) [5 marks]

Conduct a literature review to explore recent research on the use of spectroscopy data for environmental and sustainability applications. Choose at least two examples and summarise the datasets used, the challenges encountered and the use of modern regression methods in these contexts.

1.5 References

Brown, PJ (1993). *Measurement, regression and calibration*. Clarendon Press.

Ryan, T (2008). *Modern Regression Methods*, Second Edition. Wiley.

2 Computing Lyapunov exponents of chaotic systems

DR EVIATAR BACH

2.1 Overview

Chaos theory is the study of systems wherein a small change to the initial conditions leads to outcomes which diverge exponentially in time (the “butterfly effect”). Such systems are ubiquitous in nature, including the weather and the movement of the planets in our solar system.

Lyapunov exponents are a way of quantifying how chaotic a system is, and are one of the most important concepts in chaos theory. This project will involve computing Lyapunov exponents by hand for a simple system, as well as implementing computer algorithms for calculating Lyapunov exponents.

2.2 Link to global sustainability

Lyapunov exponents are important in the study of the predictability of weather and climate; i.e., how well and for how long the weather can be predicted.

2.3 Pre-requisites

This project builds on knowledge of ordinary differential equations, numerical analysis, and linear algebra.

2.4 Questions

Introduction

We start by considering one-dimensional discrete-time maps, i.e., systems of the form

$$x_{i+1}=f(x_i),$$

where i is a time index. For such a system, the Lyapunov exponent given an initial condition x_0 is

$$\lambda(x_0)=\lim_{n\rightarrow\infty}\frac{1}{n}\sum_{i=0}^{n-1}\ln|f'(x_i)|. \quad (1)$$

Having a positive Lyapunov exponent is a necessary condition for chaos.

Question 1 [6 marks]

Compute by hand the Lyapunov exponent for the map

$$x_{i+1}=f(x_i)=\begin{cases} 2x_i & \text{for } x_i < \frac{1}{2} \\ 2(1-x_i) & \text{for } x_i \geq \frac{1}{2} \end{cases}$$

for x_0 between 0 and 1.

Question 2

Consider the map

$$x_{i+1}=rx_i(1-x_i),$$

with parameter r . Despite its simplicity, this map is chaotic for some values of r .

- Approximate the Lyapunov exponent of this map at $r=3.5$ using a computer program, choosing x_0 to be a random number between 0 and 1. To test your program, note that the Lyapunov exponent at $r=2.5$ is about -0.69 . [6 marks]
- Produce a plot showing the Lyapunov exponent plotted vs. the parameter r ranging from 0 to 4. [6 marks]
- Plot a time-series of length 100 (i.e., the points $x_0, \dots, x_{100}$) for one of the values of r which does not result in chaos, and another for a value of r that does. How do these time-series differ qualitatively? [6 marks]
- For each of these values of r in d), perturb x_0 by a small amount and plot the time-series resulting from applying the map to the perturbed value, on the same plot with the unperturbed time-series. How do the non-chaotic and chaotic regimes differ? What does this imply about the predictability of chaotic systems? [6 marks]

Question 3

Lyapunov exponents can also be defined for systems of ordinary differential equations as well. Consider a system

$$\frac{d\mathbf{x}}{dt}=\mathbf{g}(\mathbf{x}),$$

where $x \in \mathbb{R}^d$. Then, this system will have d Lyapunov exponents, collectively referred to as the Lyapunov spectrum.

- a) Solve the following system numerically for $t \in [0, 100]$ using the 4th-order Runge–Kutta method with a timestep of 0.01, and plot the solution in the x – z plane:

$$\begin{aligned}\frac{dx}{dt} &= 10(y - x), \\ \frac{dy}{dt} &= x(28 - z) - y, \\ \frac{dz}{dt} &= xy - \frac{8}{3}z.\end{aligned}$$

[6 marks]

- b) Approximate numerically the maximum Lyapunov exponent for the system in a) using the algorithm described in Sprott (2015). [7 marks]
- c) Although you do not have to implement it, research an algorithm to compute the full Lyapunov spectrum of a system of ordinary differential equations and describe it. What do the other Lyapunov exponents (other than the maximum one) mean? [7 marks]

Question 4 (MA3PPNR students only) [5 marks]

Provide a brief literature review of how Lyapunov exponents have been used to quantify the predictability of a natural or engineered system, for example, the atmosphere, the movement of planets in the Solar System, or electronic circuits. What are the implications of the values of the Lyapunov exponents for the system you have chosen?

2.5 References

Strogatz, S. H. (2014). *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering* (2nd edition). CRC Press. (Available online and physically through the library. Note that the spelling “Liapunov” is used in this book.)

Sprott, J. C. (2015). *Numerical Calculation of Largest Lyapunov Exponent*. <https://sprott.physics.wisc.edu/chaos/lyapexp.htm>

3 Route-choice in offroad running

DR NICK BIGGS

3.1 Overview

In offroad running, the choice of route influences the time taken, as either varying terrain or varying hill slope will affect the runner's speed. In this project, students will investigate some simple scenarios of both types.

Specific skills used in the project include simple calculus and vector calculus, and some sort of plotting/visualisation (e.g. using MatLab or Python).

3.2 Link to global sustainability

Not applicable.

3.3 Pre-requisites

MA1CA, MA1MSP/MA2MPR, vector calculus from MA2MMP. (Note that none of the theory of functionals, Euler-Lagrange equations etc. from calculus of variations need be used.)

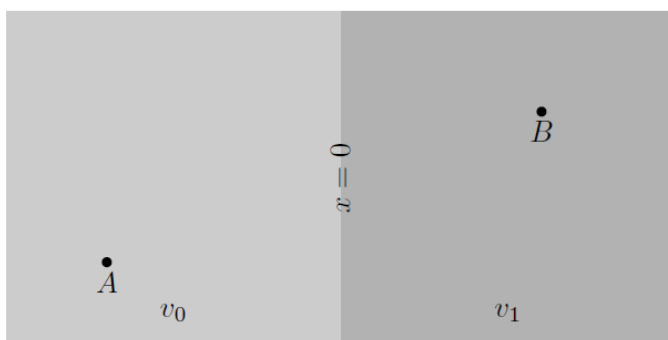
3.4 Questions

Question 1

Suppose an offroad runner wishes to cross a region of flat uniform terrain (i.e. across the plane $z=0$), on which they can run at a constant speed v_0 . The fastest route between two points in this region is evidently a straight line.

Question 1(a) [8 marks]

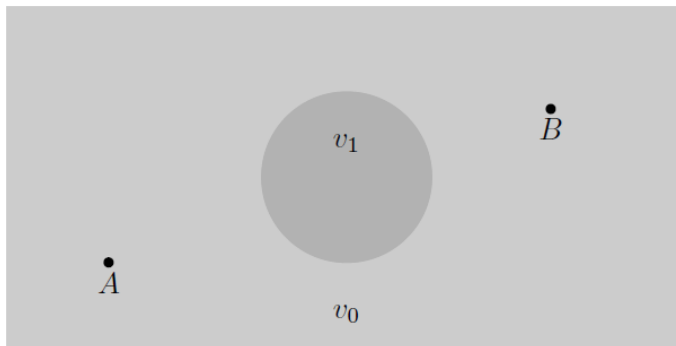
The runner now wishes to run from $A=(x_A, y_A)$ in the region $\{x < 0\}$ in which their speed is v_0 , to $B=(x_B, y_B)$ in the region $\{x > 0\}$ in which their speed is v_1 , as shown.



Investigate their quickest route, as a function of x_A, y_A, x_B, y_B, v_0 and v_1 . Use numerical minimisation to determine routes for a selection of parameter values, and plot them.

Question 1(b) [12 marks]

The runner now wishes to run from A to B , both in the region in which their speed is v_0 , the complication being that between A and B is a region in which the running speed is v_1 . See below for an example.



Choose a shape of inner region (which could be the circular shape shown, or could be something else), and investigate the fastest route in this case, again as a function of x_A, y_A, x_B, y_B, v_0 and v_1 , and possibly also parameters which define the shape of the inner region. Use numerical minimisation to determine routes for a selection of parameter values, and plot them.

Question 2

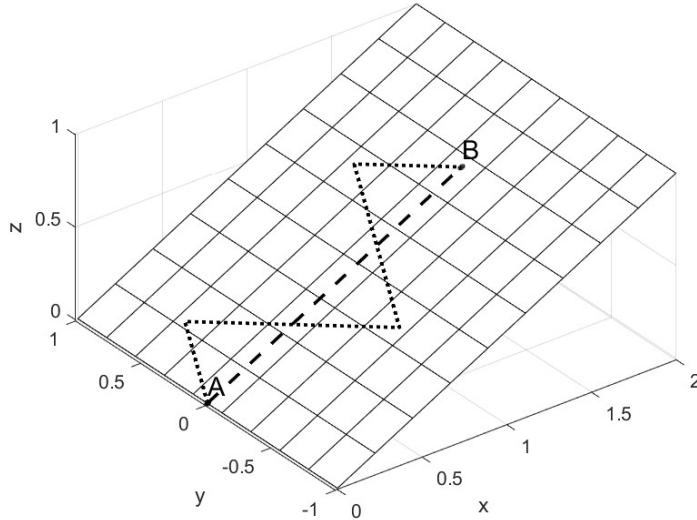
Now suppose the terrain is uniform underfoot, but that the ground is *sloped*, with the height of the ground being $z=h(x,y) \geq 0$.

Question 2(a) [5 marks]

The runner can run at speed v_0 on flat ground. Propose a speed function $v=v(g)$, in which g is the component of the gradient of h in the *direction of motion*. (Hint: increasing g above $g=0$ corresponds to increasingly steep uphill, which is hard to run up; decreasing g below $g=0$ corresponds to increasingly steep downhill, which can be hard to run down.) Plot your proposed function as a function of g , and write down an algebraic expression for it.

Question 2(b) [15 marks]

Suppose that the terrain now takes the form of the inclined plane $z=h_0x$, for $h_0>0$.



A runner wishes to get from $A=(0,0,0)$ to $B=(x_B,0,h_0x_B)$. The time taken passing along the straight line route $(y,z)=(0,h_0x)$ (the dashed line) is

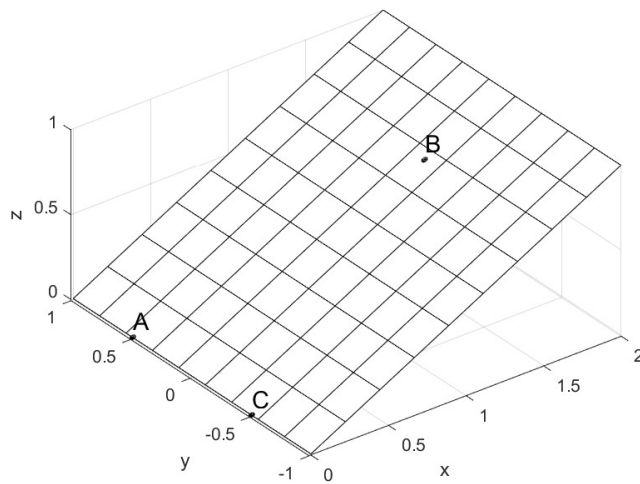
$$\frac{\text{distance}}{\text{speed}} = \frac{\sqrt{x_B^2 + (h_0x_B)^2}}{v(h_0)} = \frac{x_B\sqrt{1+h_0^2}}{v(h_0)}.$$

Investigate when a *zigzagging* route (e.g. the dotted line) is faster than this straight line route. (The idea here is that, depending on your proposed speed function, it may be quicker to run further at a faster speed, than less far at a slower speed.) Some numerical minimisation may be needed here.

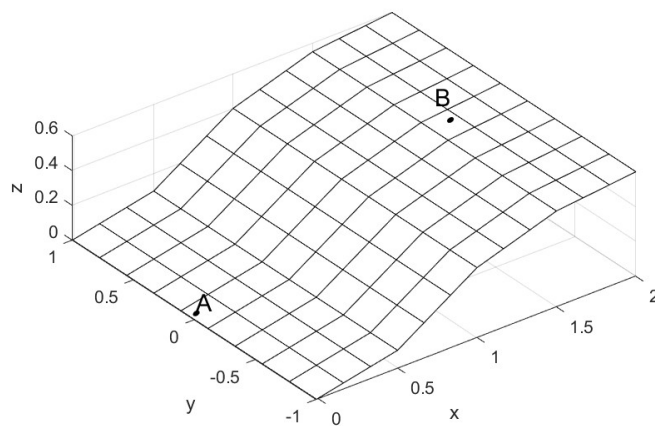
Question 2(c) [10 marks]

Investigate a further sloped hill scenario. Some options are outlined below, but others are possible - please agree any alternative scenario with Dr Biggs in advance.

1. If the terrain is steep, then running *parallel* to height contours may be hard, as the runner's feet will land unevenly. Incorporate this effect into your running speed function, and investigate its effect.
2. Determine the fastest route for an up-down run, e.g. from A to B to C as shown.



3. Investigate the fastest route for a piecewise linear hill slope $h=h(x)$, e.g. as shown.



Question 3 (MA3PPRNR students only) [5 marks]

Investigate two scenarios for Question 2(c) rather than one.

3.5 References

- Lecture notes for MA1CA, MA1MSP/MA2MPR, MA2MMP

4 Bessel functions and the Helmholtz equation

DR PETER CHAMBERLAIN

4.1 Overview

Applications involving circular geometries often involve Bessel functions. The aim of this project is to examine some Bessel functions and then use them in an application.

Subject specific skills required will involve the Part 2 Differential Equations module. In particular the method of Frobenius, separation of variables and Fourier series.

4.2 Link to global sustainability

Not applicable

4.3 Pre-requisites

Part 2 Differential Equations and some programming (in Matlab or Python) to visualise the application at the end.

4.4 Questions

Question 1 [25 marks]

We consider the differential equation

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + x^2 y = 0, \quad (0 < x < a) \quad (2)$$

in which $a > 0$.

- Show that a solution of (2) that is bounded as $x \rightarrow 0$ must be a constant multiple of

$$J_0(x) = \sum_{n=0}^{\infty} \frac{(-1)^n (x/2)^{2n}}{(n!)^2}.$$

- Let $m \in \mathbb{N}$ and generalise the calculations implied above to show that solutions of

$$x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - m^2) y = 0, \quad (0 < x < a)$$

that are bounded as $x \rightarrow 0$ are multiples of

$$J_m(x) = \sum_{n=0}^{\infty} \frac{(-1)^n (x/2)^{2n+m}}{n!(n+m)!}.$$

The function J_m , where $m+1 \in \mathbb{N}$ is called a Bessel function of the first kind of order m .

Question 2 [15 marks]

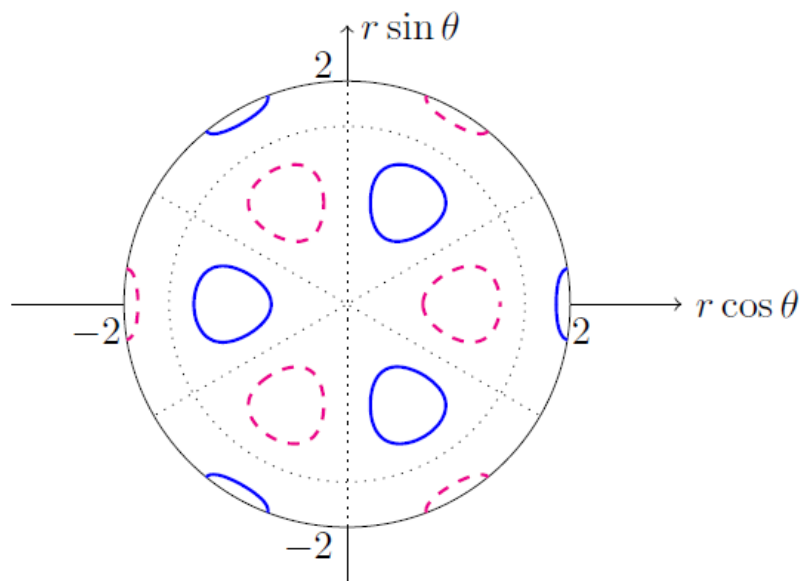
We suppose that $u=u(r,\theta)$ satisfies the Helmholtz equation inside a circle of radius $a>0$, where r and θ are plane polar coordinates. Consider the boundary value problem

$$\left. \begin{aligned} \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} + k^2 u &= 0, & (r < a) \\ u(a, \theta) &= f(\theta) & (0 \leq \theta < 2\pi) \end{aligned} \right\} \quad (3)$$

in which $k>0$. Here f is the given boundary condition and our aim is to find a solution to (3) that is

1. bounded as $r \rightarrow 0$ and
2. periodic in θ , with period equal to 2π .

Solve the boundary value problem (3) in the case where $f(\theta) = \cos(\alpha\theta)$. In the case where $\alpha=3$, $a=2$ and $k=3$ compare your results with the contour plot of the exact solution given below. The broken lines correspond to $u=-0.9$, the solid lines show $u=0.9$ and the dotted lines show where $u=0$.



Choose different values of α , a and k and illustrate the function u that arises. We will use the timetabled sessions to make sure that everybody has their own unique parameter values.

Question 3 [10 marks]

Generalise the example in Question 2 to allow for different functions f . Use the timetabled session to discuss your choice or choices here. It will be possible to choose functions f which allow any integrals that crop up to be evaluated exactly.

Question 4 (MA3PPRNR only) [5 marks]

Adapt your program so that the integrals required for the Fourier Series in f are approximated by some numerical method. There are some commands built in to Matlab and Python that can help and we can discuss these in class. Use your program to produce a plot illustrating $u(r, \theta)$ in the case where $k=1$, $a=4$ and

$$f(\theta) = \sin\left(4\left[\theta - \frac{\pi}{6}\right]\right) \exp(-(\theta - \pi)^2),$$

in which $0 \leq \theta < 2\pi$.

4.5 References

- Elementary differential equations and boundary value problems by William E. Boyce; Richard C. DiPrima. Wiley.
- Differential and integral equations by P.J. Collins. Oxford University Press.

And your Part 2 lecture notes in Differential Equations should be useful.

5 Minimization Methods

PROF SARAH DANCE

5.1 Overview

There are many mathematical problems that require the minimization of a function of several variables. For example, in machine learning, it is often the case that the neural network learns by minimizing a nonlinear loss function.

In this project the students will consider two gradient descent methods that could be used for minimization: steepest descents and conjugate gradients.

Students will be expected to carry out some mathematical analysis, computer programming, visualization of results from numerical experiments and interpretation of results from their numerical experiments.

5.2 Link to global sustainability

There are many optimization problems that contribute to global sustainability, where the best solution is found by minimizing a loss function. Some examples are

- **Computing the initial conditions for a weather forecast** (data assimilation) - combining observations of the atmosphere with numerical models that solve partial differential equations describing atmospheric flows. The loss function here measures the “distance” of the model state to the observations, using a weighted Euclidean norm, weighted by the uncertainty of each observation type. The problem is very large with typically 10s of millions of observations and only a few minutes to solve the problem. It is important that the minimization problem is solved efficiently, so that warnings of extreme weather can be issued in time for mitigating actions.
- **Renewable energy forecasting** Predicting the output of intermittent renewable energy sources like solar and wind power is important for balancing energy supply and demand, and optimizing the integration of clean energy into the power grid. Machine learning forecasting models can be trained by minimizing the difference between forecast and actual power generation.

5.3 Pre-requisites

The project builds on topics from calculus and linear algebra. There are no other prerequisites.

5.4 Questions

Introduction

We seek to minimise a real-valued function, $f: \mathbb{R}^n \rightarrow \mathbb{R}$, given that the gradient,

$$g_i = \frac{\partial f}{\partial x_i}, \quad i=1,2,\dots,N,$$

can be calculated. We are going to consider two methods that are iterative, where each iteration reduces the value of $f(\mathbf{x})$ by searching along a descent direction in the following way.

The iteration begins with a starting point $\mathbf{x}_0$, and at this point the gradient vector $\mathbf{g}$ is calculated. Then a search direction, $\mathbf{s}$, say, is chosen, that satisfies the condition $\mathbf{g} \cdot \mathbf{s} < 0$. (where the dot denotes a scalar product). It follows that if we move from $\mathbf{x}_0$ in the direction of $\mathbf{s}$, then the value of $f(\mathbf{x})$ becomes smaller initially, i.e.,

$$f(\mathbf{x}_0 + \lambda \mathbf{s}) < f(\mathbf{x}_0),$$

for some scalar value of λ . Usually we search for the value of λ that minimizes the left-hand side, λ^* . We then set the next iterate, $\mathbf{x}_1 = \mathbf{x}_0 + \lambda^* \mathbf{s}$ and another iteration is begun.

The **Steepest Descents** method uses this iteration approach with $\mathbf{s} = -\mathbf{g}$.

Question 1 Steepest Descents [15 marks]

Consider the function $f(x,y) = \frac{1}{2}(x^2 + by^2)$ for $0 < b \leq 1$.

Work out analytically

- (a) The minimum value of the function
- (b) A formula for the values of iterates with the **Steepest Descent** approach and exact line search (i.e., the best possible values of λ^*), starting from the point $(x_0, y_0) = (b, 1)$.
- (c) Make a contour plot and/or surface plot of the function values, showing the first 10 iterates from (b) for two different values of b . You should choose the values of b yourself to demonstrate interesting features of the problem.
- (d) Comment on the behaviour seen in the plot.

Question 2 Linear Conjugate Gradient Method [15 marks]

The conjugate gradient algorithm used steepest descents for its first step, and then adjusts the search direction to try to overcome the problems of steepest descents alone. The Fletcher-Reeves conjugate gradient algorithm works as follows

- Let $\mathbf{s}_0 = -\mathbf{g}$ (i.e. the negative gradient of the function at the starting point, $\mathbf{x}_0$)
- Find the value of $\lambda > 0$ that minimizes $f(\mathbf{x}_0 + \lambda \mathbf{s}_0)$, λ^* , and set $\mathbf{x}_1 = \mathbf{x}_0 + \lambda^* \mathbf{s}_0$.
- For $n=1, 2, \dots$ set $\mathbf{g}_n = f'(\mathbf{x}_n)$ and

$$\beta_n = \frac{\mathbf{g}_n \cdot \mathbf{g}_n}{\mathbf{g}_{n-1} \cdot \mathbf{g}_{n-1}}.$$

Update the search direction, $\mathbf{s}_n = -\mathbf{g}_n + \beta_n \mathbf{s}_{n-1}$ and let

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \lambda^* \mathbf{s}_n,$$

where $\lambda^* > 0$ minimizes $f(\mathbf{x}_n + \lambda \mathbf{s}_n)$. Check convergence and if not converged, repeat.

Suppose that $\mathbf{x}_0$ is the starting point and $\mathbf{x}_1$, has been obtained using steepest descents using $\mathbf{x}_0$ and the corresponding gradients, $\mathbf{g}_0$, $\mathbf{g}_1$, then

(a) Write a computer program to implement the **Conjugate Gradient Method**. You could use a fully automatic method or a human-in-the-loop method where you determine λ^* by eye at each stage. For example, you could use a simple (x,y) plot of $\phi(\lambda) = f(\mathbf{x}_n + \lambda \mathbf{s}_n)$ to help you determine the value of λ^* . (You need never determine λ^* to more than 2 significant figures). At each stage, arrange for your program to display or save the current value of $f(\mathbf{x}_n, \mathbf{y}_n)$, and iteration points $(\mathbf{x}_n, \mathbf{y}_n)$ (where n is the iteration number). You will need this data for Question 2(b)).

In your report, you do not need to include your code, but you should include some mathematical information about the method you employed.

(b) Produce a plot similar to that for Q1(c) showing the function values and as many iterations as you need to reach the minimum value of the function (within a small tolerance). Please use the same two values of b used in Q1.

(c) Compare and discuss your numerical results for the Conjugate Gradient method with the Method of Steepest Descents.

Question 3 [20 marks]

(a) Now generalize the specific minimization problem that you were solving in Q1 and Q2 to a function of the form

$$f(\mathbf{x}) = \|\mathbf{Ax} - \mathbf{b}\|^2$$

where $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{A} \in \mathbb{R}^{n \times n}$ is a symmetric, positive definite matrix. This is known as a quadratic form. Derive formulae for the search directions for both the Steepest Descent and Conjugate Gradient approaches. Do they have any special properties?

(b) Use the literature to find out convergence results for the Steepest Descents and Conjugate Gradient approaches. Discuss whether your findings in Q1 and Q2 are consistent with these results.

Question 4 Nonlinear Conjugate Gradient (MA3PPRNR students only) [5 marks]

The Fletcher-Reeves method described in Q2 can be applied to a much broader class of functions than quadratic forms.

Consider a Rosenbrock “banana” function of the form $f(x, y) = (a - x)^2 + b(y - x^2)^2$. It has a global minimum at $(x, y) = (a, a^2)$.

Taking $a = 1$ and your own choice of $b \in [80, 120]$, adapt your code from Q2 to minimize the function, starting from several different initial guesses $\mathbf{x}_0$. How does the behaviour differ from the minimization of the quadratic form in Q2? Discuss your findings, using plots to illustrate the key points. [Hint: it may be best to use a log scale for plotting function values to enable you to see clearly what is happening. This can be achieved either using built in log-scales in your chosen plotting software, or by taking a log of the function values yourself before plotting. In this case you will want to avoid taking the log of zero.)

5.5 References

Ford, William, *Numerical Linear Algebra with Applications : Using MATLAB and Octave*, Elsevier Science & Technology, 2014.

Nocedal, Jorge and Wright, Stephen J. *Numerical optimization* Springer 2006

R. Fletcher, C. M. Reeves, Function minimization by conjugate gradients, *The Computer Journal*, Volume 7, Issue 2, 1964, Pages 149–154, <https://doi.org/10.1093/comjnl/7.2.149>

6 Class groups of quadratic fields

DR CHRISTOPHER DAW

6.1 Overview

Quadratic (number) fields are the simplest non-trivial extensions of the rational numbers $\mathbb{Q}$. Just as $\mathbb{Q}$ contains the ring of integers $\mathbb{Z}$, every quadratic field K contains a ring of integers $\mathcal{O}_K$. The phenomenon on which this project is based is that the Fundamental Theorem of Arithmetic often fails in $\mathcal{O}_K$. We can measure this failure via the so-called *class number* of K , which is the cardinality of a finite group known as the *class group*. The purpose of this project will be to understand the theory behind this phenomenon and to calculate one or more class groups by hand.

6.2 Link to global sustainability

Not applicable

6.3 Pre-requisites

Students **must** have taken **MA2ALA** and feel as though they have some understanding of groups, rings, and ideals. Students who are taking MA3NTC and have enjoyed the number theory parts of the course are likely to enjoy this project, but MA3NTC is not a strict pre-requisite, though some of the material is likely to be helpful. This is a project for students who enjoy pure mathematics and are comfortable with more abstract concepts.

Students who have programming experience are welcome to use this to carry out computations, but programming skills are not required. Indeed, your supervisor has next to none himself...

6.4 Questions

In this project, you should

- (1). give your own account of the theory of class groups of quadratic fields;
- (2). provide calculations and examples of your own;
- (3). compute at least one class group by hand (I will assign you a unique group);

The idea of these notes is to introduce you to the topic. They are structured around a number of questions (or tasks) that you should complete as you go to build your knowledge. Your answers to these questions may be included in your final project. Remember to always reference any work from other sources, and to bear in mind that original work will score more marks than material copied from the literature.

The questions without marks are literature review questions. For **MA3PPNR students**, they are worth a total of 5 marks. For **MA3PPR** students they are merely prompts to guide you.

Question

Recall for yourself the definitions of a **ring** and of a **field**. Recall the definition of an **ideal** of a ring. You may wish to review your notes from MA2ALA.

Question (9 marks)

Give some examples of rings and fields, and, for each of your rings, give some ideals of that ring. (Try to give 3 rings that are not fields, 3 fields, and, for each of your rings, 3 non-trivial ideals.)

Let d be a square-free integer (that is, an integer that is not a multiple of p^2 for any prime number $p \in \mathbb{N}$ — make sure you understand this). Write $\mathbb{Q}[\sqrt{d}]$ for the ring

$$\{a + b\sqrt{d} : a, b \in \mathbb{Q}\}$$

under the standard operations of addition and multiplication. (Notice that this mimics the usual notation of $R[X]$ for the ring of polynomials over a ring R in a variable X .)

Question (3 marks)

Show that $K := \mathbb{Q}[\sqrt{d}]$ is a field. That is, K is equal to its fraction field, denoted $\mathbb{Q}(\sqrt{d})$ — notice the round brackets.

Question (2 marks)

It should be clear that K is a vector space over $\mathbb{Q}$. What is its dimension? Give the most obvious basis.

We define the **conjugate** of an element $\alpha = a + b\sqrt{d} \in K$ to be $\bar{\alpha} := a - b\sqrt{d}$.

Question (3 marks)

Show that $\alpha \mapsto \bar{\alpha}$ defines an automorphism of the field K . Show that it is the only non-trivial automorphism of K .

We define the **norm** $N_{K/\mathbb{Q}}(\alpha)$ of an element $\alpha \in K$ to be the product $\alpha\bar{\alpha}$, and we define the **trace** $\text{Tr}_{K/\mathbb{Q}}(\alpha)$ to be $\alpha + \bar{\alpha}$.

Question (5 marks)

For $\alpha, \beta \in K$ and $a \in \mathbb{Q}$, show that

- (1). $N_{K/\mathbb{Q}}(\alpha), \text{Tr}_{K/\mathbb{Q}}(\alpha) \in \mathbb{Q}$;
- (2). $N_{K/\mathbb{Q}}(\alpha\beta) = N_{K/\mathbb{Q}}(\alpha)N_{K/\mathbb{Q}}(\beta)$;
- (3). $\text{Tr}_{K/\mathbb{Q}}(\alpha + \beta) = \text{Tr}_{K/\mathbb{Q}}(\alpha) + \text{Tr}_{K/\mathbb{Q}}(\beta)$;
- (4). $N_{K/\mathbb{Q}}(\alpha) = 0 \iff \alpha = 0$;
- (5). $N_{K/\mathbb{Q}}(a) = a^2$.

An **algebraic integer** is any complex number satisfying a **monic** polynomial over $\mathbb{Z}$. Recall that a monic polynomial is a polynomial whose leading coefficient is equal to 1.

Question (10 marks)

Show that $\alpha \in K$ is an algebraic integer if and only if

$$\alpha = \begin{cases} a + b\sqrt{d} & \text{when } d \equiv 2, 3 \pmod{4} \\ a + b\left(\frac{1 + \sqrt{d}}{2}\right) & \text{when } d \equiv 1 \pmod{4}, \end{cases}$$

for some $a, b \in \mathbb{Z}$.

This is tricky, but do give it a really good try before referring to other sources. If you do refer to sources, make sure you reference them, and make sure that you understand what you write.

We denote the set of algebraic integers in K by $\mathcal{O}_K$ — that is, a curly O and a subscript K .

Question (4 marks)

Show that $\alpha \in K$ is an element of $\mathcal{O}_K$ if and only if

$$N_{K/\mathbb{Q}}(\alpha), \text{Tr}_{K/\mathbb{Q}}(\alpha) \in \mathbb{Z}.$$

Question (2 marks)

Show that $\mathcal{O}_K$ is a subring of K .

We refer to $\mathcal{O}_K$ as the **ring of integers** of K . It plays the same role in K as $\mathbb{Z}$ plays in $\mathbb{Q}$. In other words, we can perform arithmetic in $\mathcal{O}_K$ — we can factorise elements into irreducible elements.

Question

Remind yourself of the **Fundamental Theorem of Arithmetic**.

Question

Remind yourself of the definition of the **units** $R^\times$ of a ring R . Recall some examples.

Question (2 marks)

Show that $\alpha \in \mathcal{O}_K$ is a unit if and only if $N_{K/\mathbb{Q}}(\alpha) = \pm 1$.

We say that an element $\alpha \in \mathcal{O}_K$ is **irreducible** if any factorisation $\alpha = \beta\gamma$ requires β or γ (or both) is a unit.

Question (2 marks)

Show that $\alpha \in \mathcal{O}_K$ is irreducible if $N_{K/\mathbb{Q}}(\alpha) \in \mathbb{Z}$ is prime (positive or negative).

Every non-zero non-unit in $\mathcal{O}_K$ is a product of irreducible elements of $\mathcal{O}_K$. You may wish to write a full proof of this fact.

Question (2 marks)

By considering different choices of d and different elements of $\mathcal{O}_K$, show that the factorisation of an element $\alpha \in \mathcal{O}_K$ need not be unique. That is, we may have $\alpha = \beta\gamma = \zeta\eta$ with $\beta, \gamma, \zeta, \eta$ all non-units, no two of which are equal up to multiplication by a unit.

It is this phenomenon on which this project is based.

It turns out that, while unique factorisation doesn't necessarily hold for elements, it does hold for ideals of $\mathcal{O}_K$.

For two ideals $\mathfrak{a}, \mathfrak{b} \subset \mathcal{O}_K$, we define the product $\mathfrak{a}\mathfrak{b}$ to be the set of sums $\sum_{i=1}^n a_i b_i$ for some $n \in \mathbb{N}$, $a_i \in \mathfrak{a}$, and $b_i \in \mathfrak{b}$.

Question (2 marks)

Show that $\mathfrak{a}\mathfrak{b}$ is an ideal of $\mathcal{O}_K$.

We say that an ideal $\mathfrak{b}$ divides $\mathfrak{a}$ (and we write $\mathfrak{b}|\mathfrak{a}$) if $\mathfrak{a}=\mathfrak{b}\mathfrak{c}$ for some ideal $\mathfrak{c}$.

For an ideal $\mathfrak{a}$ of $\mathcal{O}_K$, we denote by $\bar{\mathfrak{a}}$ the ideal consisting of the elements $\bar{a}$ for $a \in \mathfrak{a}$. The ideal $\mathfrak{a}\bar{\mathfrak{a}}$ is generated by elements in $\mathbb{Z}$, hence it is principal. We define the norm $N_{K/\mathbb{Q}}(\mathfrak{a})$ to be the positive integer n such that the product ideal $\mathfrak{a}\bar{\mathfrak{a}}$ is equal to the principal ideal (n) generated by n .

Question (2 marks)

Show that, for nonzero ideals $\mathfrak{a}, \mathfrak{b} \subset \mathcal{O}_K$, we have

$$N_{K/\mathbb{Q}}(\mathfrak{a}\mathfrak{b})=N_{K/\mathbb{Q}}(\mathfrak{a})N_{K/\mathbb{Q}}(\mathfrak{b}).$$

Question (2 marks)

For an element $\alpha \in \mathcal{O}_K$, check that the norm $N_{K/\mathbb{Q}}((\alpha))$ of a principal ideal (α) is equal to $|N_{K/\mathbb{Q}}(\alpha)|$.

We say that an ideal $\mathfrak{p} \subsetneq \mathcal{O}_K$ is a **prime** ideal if, for any $a, b \in \mathcal{O}_K$,

$$ab \in \mathfrak{p} \implies a \in \mathfrak{p} \text{ or } b \in \mathfrak{p}.$$

For any prime ideal $\mathfrak{p}$ and ideals $\mathfrak{a}, \mathfrak{b}$,

$$\mathfrak{p}|\mathfrak{a}\mathfrak{b} \implies \mathfrak{p}|\mathfrak{a} \text{ or } \mathfrak{p}|\mathfrak{b}.$$

Notice how this mimics the property of prime numbers in $\mathbb{Z}$.

Every nonzero proper ideal of $\mathcal{O}_K$ admits a factorisation into a product of prime ideals and this factorisation is unique up to the ordering of the factors. You may wish to write a full proof of this fact, imitating the proof of the Fundamental Theorem of Arithmetic.

On the other hand, the ring $\mathcal{O}_K$ has unique factorisation (for elements) if and only if every ideal of $\mathcal{O}_K$ is principal. We will quantify the failure of unique factorisation in $\mathcal{O}_K$ by an object known as the **class group** of K .

Question

Review the notions of a fractional ideal and a principal fractional ideal of K . These are slight generalisations of the notions of an ideal and a principal ideal of $\mathcal{O}_K$. Have a look at the Introduction to [3].

Under multiplication (as above), the fractional ideals of K form a group J_K (for which $\mathcal{O}_K$ is the identity). The principal fractional ideals constitute a subgroup P_K of J_K , and the quotient group $C_K := J_K/P_K$ is finite.

We refer to C_K as the **class group** of K .

Question

Review the facts above. See Section 2 of [3].

You will be given a square-free integer d and your task will be to calculate the class group of $\mathbb{Q}(\sqrt{d})$ by hand. Depending on your progress, you may wish to calculate more than one class group, but this is not expected. You can choose other d 's yourself or I can give you another.

Question

Familiarise yourself with the Minkowski bound. This will play a vital role in your calculations.

Don't worry, there are plenty of examples out there, but you must tell me if, and as soon as, you find your particular example worked out somewhere else.

You should write your calculations up in detail, explaining all of your working, and all of the background material necessary to understand them. Computer packages are allowed. In particular, you may use them to check what the class group is before you set about your calculations. The calculation itself must be by hand, however, and you should include any computer assistance (code etc.) as an appendix in your final project.

Once you've worked through the above, I would recommend working through Example 2.2 in [1] — this will show you the method to calculate the class group. One of the main results Conrad uses is Theorem 8.3 from [2]. Make sure you pay attention to the value of your d modulo 4!

Good luck!

6.5 References

The primary sources for this project are some unpublished notes by the brilliant Kevin Conrad, all available on his webpage <https://kconrad.math.uconn.edu/blurbs>. To access the resources below, append to the URL /gradnumthy/ and then the specific suffix below. Of course, you should seek other sources as and where necessary to help you understand these.

[1] Conrad, K., Class Group Calculations, suffix:classgpex.pdf

[2] Conrad, K., Factoring in Quadratic Fields, suffix:quadraticgrad.pdf

[3] Conrad, K., Ideal Classes and the Kronecker Bound, suffix:classgroupKronecker.pdf

7 Mathematics of Liquid Crystals

DR PATRICK ILG

7.1 Overview

Liquid Crystals provide a unique state of matter that combines liquid aspects due to positional disorder with orientational ordering. Computer monitors, digital cameras and mobile phones often use liquid-crystal displays. Mathematics not only provides useful tools for solving problems in Liquid Crystal research, but also faces interesting challenges.

Most projects require a certain amount of programming and analysing results. Visualisation of results is generally expected.

7.2 Link to global sustainability

Liquid crystals and liquid crystal elastomers are currently studied intensively to explore their use as sustainable responsive materials to transform industry, buildings and medicine.

7.3 Pre-requisites

Students may find material from one or more of the modules MA2NAN, MA2MOD, MA2MPH, MA3MP and MA3APS to be useful for this project, but it is not a requirement that students have taken them.

7.4 Questions

Question 1 [8 marks]

Give a basic introduction to Liquid Crystals. In no more than two pages, explain what Liquid Crystals are, why they are interesting for applications and why they are interesting for mathematics.

Question 2 [42 marks]

Students can choose to work on one of the four topics below and address the corresponding questions. Subquestion iv can be used to demonstrate understanding and independence.

MA3PPRNR students also need to work on one subquestion of a second topic for an additional 5 marks.

Questions 2.1: Minimisers and Equilibrium States of Liquid Crystals

- i. Define and explain the Oseen-Frank, Landau-de Gennes, and Onsager free energy functionals of Liquid Crystals.
- ii. Define equilibrium states as minimizers of these free energy functionals and explain corresponding variational principles.
- iii. Discuss existence and properties of minimisers for spatially homogeneous systems for the Landau-de Gennes and Onsager free energies.
- iv. State, prove and illustrate at least one result on minimisers for spatially inhomogeneous systems.

Questions 2.2: Orientational Order in Liquid Crystals

- i. Define and explain the orientational probability density $f(\mathbf{p})$ with $\mathbf{p}$ an element of the three-dimensional unit sphere and the 3×3 matrix $\mathbf{Q} = E(\mathbf{p} \otimes \mathbf{p}) - (1/3)\mathbf{I}$, where E denotes the expectation value.
- ii. Discuss uniaxial and biaxial order in terms of f and $\mathbf{Q}$. Use examples of f to demonstrate your results.
- iii. Explain and discuss the closure problem of expressing $E(\mathbf{p} \otimes \mathbf{p} \otimes \mathbf{p} \otimes \mathbf{p})$ in terms of $\mathbf{Q}$. In which areas is the closure problem relevant?
- iv. Test several closure approximations that have been suggested in the literature. Use your own examples of different f . Can you suggest your own closure approximation that works particularly well for some special cases?

Questions 2.3: Twisted nematic cell

- i. Explain twisted nematic cells and their use. Formulate the Oseen-Frank free energy functional for one-dimensional orientation profiles $\mathbf{n} = (\sin \theta(x), \cos \theta(x), 0)^T$.
- ii. Find the differential equation that the minimiser $\theta(x)$ must satisfy.

- iii. Solve the boundary value problem to find minimisers numerically for the twisted nematic cell and critically analyse your result for the twist profile.
- iv. Extend your study in iii by introducing an obstacle in the cell. You may use different shapes and/or different boundary conditions on the obstacle.

Questions 2.4: Lattice models of Liquid Crystals

- i. Review some lattice models of Liquid Crystals and discuss their properties.
- ii. Implement the energy function of the two-dimensional Lebwohl-Lasher model on a square lattice (use e.g. Matlab, python). Specify to parallel anchoring on all four sides of a square domain of length L .
- iii. Find the corresponding minimum energy states using the steepest descent method. Is the minimum unique?
- iv. Extend your study in iii e.g. by changing the boundary conditions, using a different minimisation method, or allowing three-dimensional orientations.

7.5 References

1. J.M. Ball. Mathematics and Liquid Crystals. Mol. Cryst. Liq. Cryst. 647, 1-27 (2017). <http://dx.doi.org/10.1080/15421406.2017.1289425>
2. S. Chandrasekhar. Liquid Crystals. Cambridge University Press 1992.
3. P. Palffy-Muhoray. Lecture slides on Liquid Crystals. Boulder School 2011. https://boulderschool.yale.edu/sites/default/files/files/Boulder_LC_Talk_Lecture_1.pdf

8 Introduction to Differential Geometry

Curves in $\mathbb{R}^3$

DR SUGATA MONDAL

8.1 Overview

The purpose of this project is to provide an introduction to differential geometry in the form of curves in $\mathbb{R}^3$.

A *smooth* curve in $\mathbb{R}^3$ is a function $\gamma: I \rightarrow \mathbb{R}^3$, where I is either an open interval (a, b) with $a < b$, or a closed interval $[a, b]$ with $a < b$ and $a, b \in \mathbb{R}$, that is k times differentiable for any $k \geq 1$. In terms of the coordinates in $\mathbb{R}^3$ we may write

$$\gamma(t) = (\gamma_1(t), \gamma_2(t), \gamma_3(t)).$$

Therefore, γ is smooth if and only if each $\gamma_i: I \rightarrow \mathbb{R}$ is k times differentiable for any $k \geq 1$. We say γ is regular if $\gamma'(t) = (\gamma'_1(t), \gamma'_2(t), \gamma'_3(t)) \neq (0, 0, 0)$. Recall that $\gamma'(t)$ is called the *velocity* vector of the curve γ at t .

The project will begin by recalling the notion of length of a parametrized curve. This will provide the set up for arclength parametrization, Frenet frame, curvature and torsion of a curve. Using these notions one can classify curves of constant curvature. The notion of curvature and torsion can also be used to determine when two curves differ by a rigid motion (the fundamental theorem of curve theory). In the end we will see some global invariants of curves e.g. Crofton's formula, Hopf Umlaufsatz. The material can be found in Chapter 1 of [S]. A mature student can also consult Chapter 1 and 2 of [K].

8.2 Link to global sustainability

Not applicable.

8.3 Pre-requisites

Students must take MA2RAT. Good understanding of functions of several variables and linear algebra is necessary.

Questions

In this project, you should

1. give your own examples of smooth regular curves,

2. give your own account of the fundamental theorem of curve theory
3. find explicit examples of curves and do clear computations to verify Crofton's formula and Hopf Umlaufsatz.

Question 1 [MA3PPRNU students only] Recall the notion of length of a curve. Take an example and compute it's length. [5 marks]

Question 2 Recall the notions of curvature and torsion of a curve. Take an example and compute these quantities. [12 marks]

Question 3 Take a curve $\gamma: I \rightarrow \mathbb{R}^3$, and a 3×3 orthogonal matrix A with real entries. Consider the curve $\gamma_A: I \rightarrow \mathbb{R}^3$

$$\gamma_A(t) = A\gamma(t) = A(\gamma_1(t), \gamma_2(t), \gamma_3(t))^t.$$

Check, by doing explicit computation that the curvature and the torsion of the two curves γ, γ_A are the same at every point. [15 marks]

Question 4

Find a proof of Crofton's formula following question 12 in Exercises 1.3 in [S]. [13 marks]

Question 5

Take an explicit example of a plane curve and verify Hopf Umlaufsatz. [10 marks]

Bibliography

- [K] Wilhelm Klingenberg; *A Course in Differential Geometry*. Translated from the German by David Hoffman Grad. Texts in Math., Vol. 51, Springer-Verlag, New York-Heidelberg, 1978. xii+178 pp. ISBN:0-387-90255-4
- [S] Theodore Shifrin; *Differential Geometry: A First Course in Curves and Surfaces*. available online at <https://math.franklin.uga.edu/sites/default/files/inline-files/ShifrinDiffGeo.pdf>

9 Expectation-Maximization (EM) algorithm and some applications

DR ABISHEK PAL MAJUMDER

9.1 Overview

Maximum Likelihood Estimation (MLE) is a widely used method in parametric statistical inference for estimating unknown parameters. However, its application becomes challenging in the presence of missing data or unobserved latent variables, as likelihood-based inference may become infeasible. The Expectation-Maximization (EM) algorithm addresses these challenges by providing an iterative framework to estimate unknown parameters even when data is incomplete or partially observed. This project focuses on understanding how the EM-algorithm works and its application in two key contexts: handling missing data and solving Gaussian cluster identification problems and in another example related to finding the blood-group genotypic probabilities from the frequency of the blood-group data. Clustering is a fundamental task in data analysis with applications in various domains such as biology, marketing, and image processing. However, the effectiveness of traditional clustering methods is significantly reduced when datasets contain missing values. The EM-algorithm, integrated with Gaussian Mixture Models (GMMs), offers a probabilistic approach to tackle these issues by iteratively estimating missing values and optimizing clustering parameters.

The primary goal of this project is to implement the EM-algorithm for GMM-based clustering and evaluate its performance on both synthetic and real-world datasets. Experiments will assess the robustness of the algorithm across different scenarios, including varying proportions of missing data and complex cluster structures. Through these efforts, the project will provide insights into the EM algorithm's strengths and limitations, highlighting its potential for practical applications in real-world data analysis tasks involving incomplete or noisy datasets.

9.2 Link to global sustainability

Not applicable.

9.3 Pre-requisites

The project builds on topics of *Statistical inference* and *multi-variate statistics* including maximum likelihood and multi-variate normals etc. It is good to have linear models (**ST2LM/ST2LMD**) and machine learning (**ST3MML**). Experience of programming is necessary (e.g. from **MA1MSP**, **CS1PC20** or from elsewhere).

9.4 Questions :

There are two applications where these EM algorithms can be effectively applied. One is in contexts of Gaussian clustering problem where latent values are cluster identification id of each observed sample (see [[Ref. 2](#)]) and the second one is in context of finding the genotypic blood-group probabilities (p_A, p_B, p_O) from the frequency of blood groups (A, B, AB, O) in a population (see the first example of [[Ref. 1](#)]). In order to find the MLE of the unknown parameter one needs to first write down the complete likelihood including all the latent variables/missing data and then write down the estimates of the unknown parameters after solving the likelihood equations (**M-step**). Since these estimators contain the values of the latent unobserved variables they need to be replaced by the conditional expectation of these latent states given the parameter estimates which are usually obtained from the last iterative states (**E-step**). Furthermore one needs to comment on the convergence diagnostics, such as how to select the initial values of the parameters so that consistency of the parameter estimates to the true value can be ensured. Following are the three questions that need to be addressed concretely for the aforementioned project applications (or any other related applications that can be found in the relevant references of [[Ref. 1](#)], [[Ref. 2](#)] or in [[Ref. 3](#)]) and they are following.

Question 0 (For MA3PPNR students only) [5 marks]

Write a coherent literature review.

Question 1 How to build the E-step? [15 marks]

For the selected application, after mentioning the problem, clearly describe the E-step and how this conditional expectation of the latent variables or any other features of the conditional distribution of the unobserved variables given the parameter can be established and computed from the parameter estimates of the past iterative step.

Question 2 How to make the M-step? [15 marks]

Clearly describe the M-step and how to find the estimate of the parameters from the complete data likelihood, and how to solve them depending on the contexts of the applications.

Question 3: How does the EM-Algorithm work? [20 marks]

In this part, questions like how to select initial parameter estimates from the data so that consistency of the final parameter estimates to the true value can be ensured, will be addressed.

To understand how convergence diagnostics of this algorithm works, one may look at [[Ref. 3](#)] or any other related texts.

Bibliography

- [**Ref. 1**] HAUGH, M.(2015). IEOR E4570 course note: Machine Learning for OR&FE. [Link](#).
- [**Ref. 2**] EXPECTATION–MAXIMIZATION ALGORITHM. Wikipedia page. [Link](#).
- [**Ref. 3**] HASTIE, T. TIBSHIRANI, R. FRIEDMAN, J.(2001). (Chapter 8.5) The EM algorithm: *The Elements of Statistical Learning*. New York: Springer. pp. 236–243. ISBN 978-0-387-95284-0. [Link](#).
- [**Ref. 4**] WU, C. F. JEFF(1983). On the Convergence Properties of the EM Algorithm. *Annals of Statistics*. 11 (1): 95–103.

10 Time series modelling

DR KAREN POULTER

10.1 Overview

A time series is a set of measurements made at different (usually equally spaced) points in time. Time series arise in many situations, and it is common to want to describe features of the series (like trend and seasonality), and predict likely future observations. These aims can be met by fitting a time series model to the data. As with all statistical modelling, it is important to assess the fit of a time series model and identify whether the modelling assumptions are reasonable. Time series modelling therefore consists of (a) identifying which type of model is appropriate, (b) estimating the parameters of the model, (c) checking the fit of the model and typically (d) using the model to forecast future observations.

This project will involve reading about different methods for modelling data in the time domain, and using a statistical computing package (SAS, R or something else) to fit and assess models for two different series of data. Different students will be given a different set of data to analyse (available datasets consist of economic data, energy data, crime data, TV viewing data, environmental data and health data).

Students can use SAS, R or another statistical computing package of their choice to fit and assess the models.

10.2 Link to global sustainability

There is a small number of environmental/energy datasets which can be analysed as part of this project.

10.3 Pre-requisites

The project builds on topics from linear models (ST2LM / ST2LMD) and so students must have studied one of these modules, or a similar module. There is no need to have completed the forecasting topic in the MA3PPR poster project before undertaking this project.

10.4 Questions

Question 1 [18 marks]

Using information you find in the literature, describe the components that make up an ARMA time series model, and explain how the parameter estimates are obtained for such a model. What is the difference between an ARMA and an ARIMA model? How does a S-ARIMA model differ from ARIMA? When would each model be used? Make sure that you provide sufficient theory (including equations) and detail to answer this question for someone who is unfamiliar with the subject of time series.

Question 2 [8 marks]

Describe the difference between the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF), and show examples of how they can be used to identify which terms are needed in an ARMA, ARIMA or S-ARIMA model.

Question 3 [8 marks]

Describe how we can assess the fit of a time series model. (Hint: use your study of the literature to explain the role of residuals and to describe relevant hypothesis tests.)

Question 4 [16 marks]

Fully analyse two series of data from your given dataset using time series modelling. For each series, fully explain how you determined your final chosen time series model(s). Use the model(s) to forecast future values.

Question 5 for MA3PPNR students only [5 marks]

Using information you find in the literature, describe the topic of multivariate time series. In particular you should explain what cross-correlation is, and you should describe Vector Autoregressive (VAR) and vector ARMA models. Also, identify in the literature one example of when VAR or VARMA models have been used to analyse data, and explain this example (you do not need to provide full mathematical details of the example you find).

10.5 References

Chatfield, C. and Xing, H. (2019) The analysis of time series: an introduction with R. (Seventh edition). CRC press. Available to read online at <http://ebook-central.proquest.com/lib/reading/detail.action?docID=5760899>

PennState: Statistics Online Courses, (n.d.), Applied time series analysis. The Pennsylvania State University. Available at: <https://online.stat.psu.edu/stat510/>

11 Modelling time-to-event data

PROF SUE TODD

11.1 Overview

In many everyday situations data arise which capture the length of time until the occurrence of a particular event. Examples can be readily found in the fields of medicine, engineering and sport. In medicine, examples include time to relapse in patients with psychosis, or time to remission in patients with a mental health condition. In engineering, examples are time to mechanical failure on a car or the product lifetime of a smartphone. In sport, examples could include time to crashing in a go-kart race or runs scored before getting out in cricket. Survival analysis or time-to-event analysis is the study of such data.

Modelling survival data allows researchers to explore the factors which influence occurrence of the event of interest. There are various approaches to fitting an appropriate model, all of which have in common the fundamental aspects of statistical modelling which involve developing candidate models via an iterative process, checking the fit of the models, and finally drawing appropriate inferences.

The aim of this project is to investigate the way in which survival data can be modelled. You will research the various modelling approaches and then apply them to real survival data from a particular application area. You may use either SAS or R to conduct the analyses.

11.2 Link to global sustainability

Not applicable.

11.3 Pre-requisites

This work will build on various topics covered in Linear Models and ST2LMD/ST2LM is a pre-requisite.

11.4 Questions

Question 1 [15 marks]

Explore the literature to discover the difference between the semi-parametric and parametric approaches to modelling time-to-event data. Describe examples of models which belong to these two categories. Note: You do not need to consider the AFT (Accelerated Failure Time model). You should include some relevant theory and equations in your description and enough detail and definitions in the answer to this question for someone who is unfamiliar with survival analysis.

Question 2 [15 marks]

Describe how to approach fitting a model to survival data to enable identification of the important factors affecting the event of interest. Include in your description the processes involved in model selection and assessing model fit. Explain how to interpret the parameters from a model.

Question 3 [20 marks]

For your dataset select, giving reasons for your choice, a candidate set of 8-10 explanatory variables to use for building a survival model. Using appropriate model fitting approaches, fit both a semi-parametric model and one appropriate parametric model to your data, identifying which of your candidate explanatory variables are important in influencing the event of interest. Give details of the approach you take and interpret the parameters from your final models.

Question 4 (MA3PPRNR students only) [5 marks]

Describe the Accelerated Failure Time model, including mathematical equations as appropriate. Explain how this model relates to the models which you have been researching and applying in Questions 1-3.

11.5 References

Collett, D: Modelling survival data in medical research, 3rd edition (2013)

Lee, E.T. and Wang J.W. Statistical methods for survival data analysis, 4th edition (2013)

12 Numerical Simulation Methods and Their Applications

DR ZUOWEI WANG

12.1 Overview

Computer simulations are mathematical modelling techniques that are widely applied in studying various scientific, engineering, economic and social science subjects. These algorithms are based on the principles of classical and/or statistical mechanics, probability and statistics, numerical solutions of ordinary and stochastic differential equations, statistical sampling methods, etc. Two classes of simulation techniques, namely Monte Carlo (MC) and molecular dynamics (MD) simulation methods, have been widely used in studying particle-based systems, such simple liquids, polymers and biomolecules. In this project, students can choose to work on one of the MC or MD simulation methods and its applications in studying the structural and/or dynamic properties of one representative particle-based system.

It will be helpful but not necessary if the students have some programming experience in using any language, such as Matlab, Python, C/C++ etc.

12.2 Link to global sustainability

The simulation methods developed in this project can be applied to study the structural, dynamical and mechanical properties of various soft matter systems, such as associative polymers, biopolymers (proteins, peptides, and DNA) and bio-mimicking materials. The simulation results and the insights gained from them can provide guidance for designing novel functional materials which are recyclable, reprocessible and environment-friendly, as well as highly efficient antimicrobial and antiviral drugs, and so contribute to global sustainability.

12.3 Pre-requisites

Students may find materials from one or more of the modules MA2NAN, MA2MOD, MA2MPH, MA3MP and MA3APS to be useful for this project, but it is not a requirement that students have taken them.

12.4 Questions

Students can choose to work on one of the following topics:

- Monte Carlo Methods and Applications in Particle-based Systems (e.g., Metropolis method for simulating Lennard-Jones liquid or single bead-spring chain);
- Molecular Dynamics and Finite Difference Methods (i. e., finite difference methods for numerical solutions of Newton's equation of motion, MD simulation of Lennard-Jones liquid or single bead-spring chain);
- Molecular Dynamics Methods in Different Ensembles (e.g., numerical solutions of Newton's equation of motion, MD simulations of Lennard-Jone liquids in microcanonical (NVE) and canonical (NVT) ensembles)

In each project, students are expected to address the following questions:

Question 1 [5 marks]

Write a brief review about the development of computer simulation methods, starting from the Monte Carlo method developed in the Manhattan Project to later development of MC and MD methods with the enhanced computational power.

Question 2 [10 marks]

For students working on MC methods, i) describe one or two pseudo random number generators (PRNG) and give examples to show how the PRNG(s) work; ii) describe the simple and importance sampling methods and, if having time, give examples on using these sampling methods for numerical integrations, such as calculating the value of π ; iii) discuss the statistical mechanics background of Monte Carlo methods; iv) derive the Metropolis method by using the concepts of importance sampling and detailed balance.

For students working on MD methods: i) discuss the classical mechanics background of equations of motion; ii) a) If working on MD with finite difference methods, derive two or more finite difference methods for numerical solutions of Newton's equation of motion (2nd order ODE) using Taylor expansion; b) If working on MD in different ensembles, derive at least one finite difference method and discuss at least one thermostat for maintaining constant temperature in the canonical ensemble; iii) discuss the orders of accuracy in calculating the positions and velocities of the particles by using different finite difference methods.

Question 3 [2 marks]

Describe and sketch the Lennard-Jones potential and its application in modelling simple liquids.

Describe the concept of periodic boundary conditions (PBC) as used in simulating particle-based systems and insert a figure to demonstrate how PBC works.

Question 4 [10 marks]

Describe the steps for implementing the selected MC or MD methods into a simulation code for studying simple liquid or single bead-spring chain, including initial configuration generation, loops for generating the moving trajectories of the particles and post-simulation data analysis, etc.

Develop your own simulation code using any programming language OR download and describe a simulation code found in open resources. You must clearly state the origin of the simulation code, either self-developed or downloaded from open resource (give the link or citation of the resource).

Question 5 [20 marks]

Apply the developed or downloaded code to simulate a simple Lennard-Jones liquid with 50-200 particles or a single bead-spring chain with 20-50 beads and demonstrate the equilibration process of the system by calculating the total energy of the system over the simulation steps and showing the results in plot(s).

Perform numerical analysis of some structural and dynamical properties of the system, such as the radial distribution function of particles in the simple LJ liquids or the end-to-end distance and radius of gyration of the single bead-spring chain, the mean-squared displacements and diffusion coefficients of particles in the system, etc.

If not running a simulation code, find and discuss simulation results on the structural properties of LJ liquids or bead-spring polymer chains in literature, and derive the mathematical formulae for predicting these structural and dynamical properties. You must clearly state the resource of these results.

Question 6 [3 marks]

Draw conclusions and give perspectives of the study.

Question 7 [5 marks] For MA3PPRNR students (NUIST students who are studying in Reading),

Write a literature review about theoretical calculations of the conformational and/or dynamical properties of ideal polymer chains (e. g., see Ref. 4 below). This review can be integrated with your answer to Question 5.

12.5 References

1. Computer Simulation of Liquids, M. P. Allen and D. J. Tildesley, Oxford Science Publications, 1987;
2. Understanding Molecular Simulations: From Algorithms to Applications, Daan Frenkel and Berend Smit, Academic Press, 2001;
3. The Art of Molecular Dynamics Simulation, D. C. Rapport, Cambridge University Press, 2004.
4. Polymer Physics, Michael Rubinstein and Ralph H. Colby, Oxford University Press, 2003.
5. Any other books or online resources about computer simulations.

13 Entropy in information theory

DR JEROEN WOUTERS

13.1 Overview

Information theory is without doubt one of the cornerstones of our digital society. In his seminal 1948 paper “A Mathematical Theory of Communication” (one of the most influential and cited scientific papers of all time), Claude Shannon introduced the central concept of information entropy.

Entropy and related concepts, such as mutual information, play an important role in many applications we now take for granted: data compression, electronic communication, image processing, computer linguistics, and much more.

In this project you will study the entropy as a mathematical concept, its properties and explore its applications.

You will be expected to carry out some mathematical analysis, computer programming, visualization of results from numerical experiments and interpretation of results from numerical experiments. Programming and visualization can be performed in any mainstream computer language you are comfortable with (including Python, R, C++, Matlab or Julia).

13.2 Link to global sustainability

Concepts from information theory appear in many contexts in the geosciences. Examples include the use of mutual information to process satellite imagery, or information flow to investigate causal relationships between climate modes. Such topics can be investigated as part of Question 5 below.

13.3 Pre-requisites

Familiarity with probability theory is required. An understanding of the basic concept of (discrete) probability distributions (as covered in ST1PS) will be sufficient. Students may find the further material on probability theory from the module MA2PT1 or ST2PST useful, but it is not a requirement that students have taken it.

13.4 Questions

Question 1. Give the definition of Shannon entropy. State, prove and discuss one or more of its most important mathematical properties. Plot the Shannon entropy for the Bernoulli family of distributions as a function of its parameter and discuss the result. [8 points]

Question 2. State and derive the Asymptotic Equipartition Property. Discuss its relevance in the context of data compression. [10 points]

Question 3. Discuss Huffman coding and its expected codeword length. [10 points]

Question 4. Download the digital text of any publicly available book, for example from the Project Gutenberg.

Load the file using a programming language, and compute the number of times that each character x_i appears. Estimate the probability $p(x_i)$ of each character.

Compute the entropy per character. Multiply the entropy per character by the number of characters in the book, to obtain the overall entropy of the book. Compare this to the file size of the text and of a compressed zip-file containing only the same text file. [8 points]

Question 5. Discuss one or more applications of information theory, such as communications engineering, image processing, computer linguistics, machine learning, compression algorithms, error-correcting codes, or statistical physics. *University of Reading students* should pick one field of application to discuss. *NUIST students* should cover two different types of application.

(Reading students) [14 points]

(NUIST students) [19 points]

13.5 References

1. Cover & Thomas “*Elements of Information Theory*” John Wiley & Sons, Inc, 1991
2. Pinkard, H., Waller, L., 2022. A visual introduction to information theory. <https://doi.org/10.48550/arXiv.2206.07867>
3. Shuh “*Information Theory for Data Science*” Norwell, Now Publishers 2023
4. MacKay “Information Theory, Inference, and Learning Algorithms” Cambridge University Press 2003. (<http://www.inference.org.uk/itprnn/book.pdf>)
5. Shannon, C.E., 1948. A mathematical theory of communication. The Bell System Technical Journal 27, 623–656. <https://doi.org/10.1002/j.1538-7305.1948.tb00917.x>